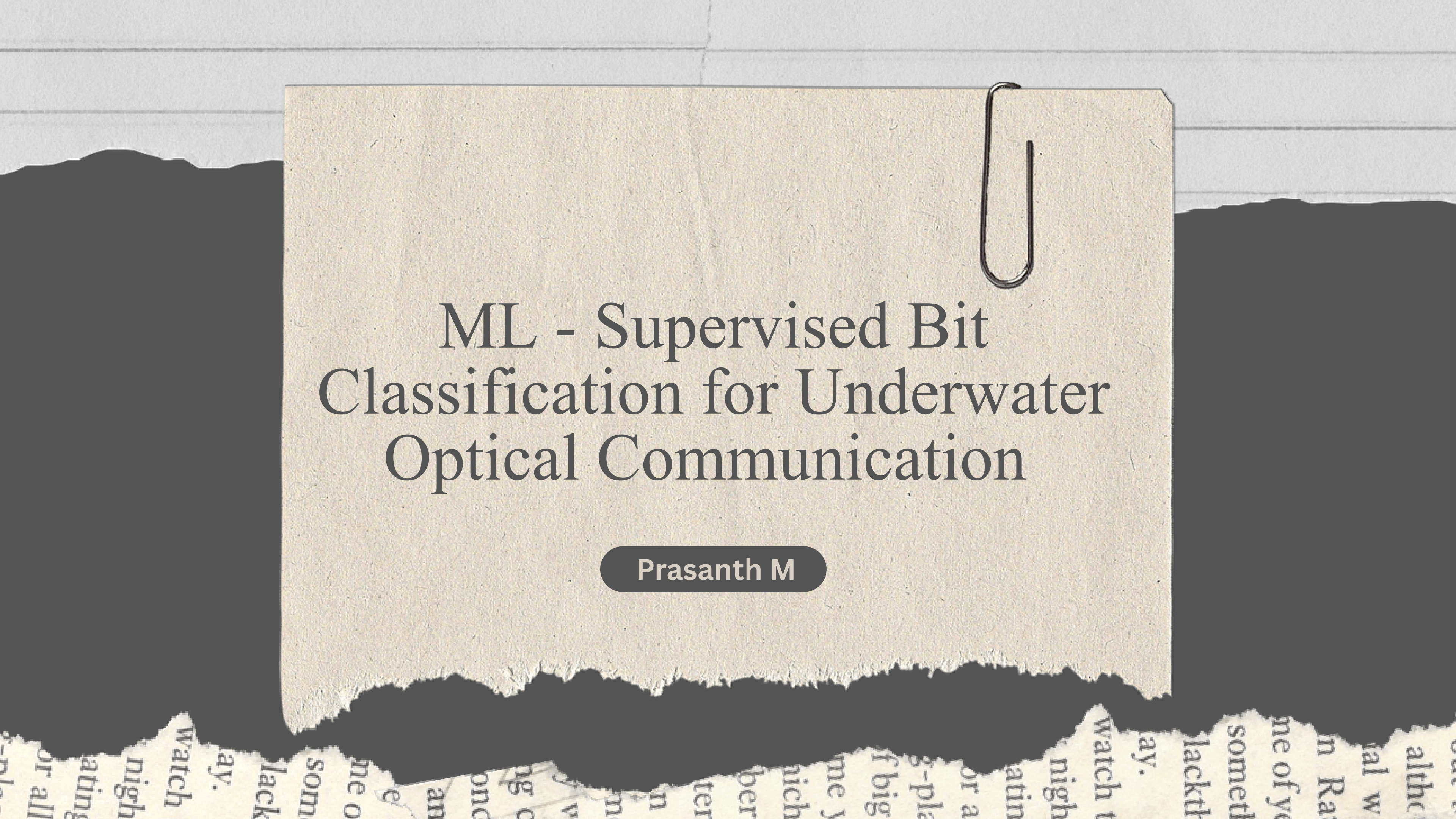




ML - Supervised Bit Classification for Underwater Optical Communication

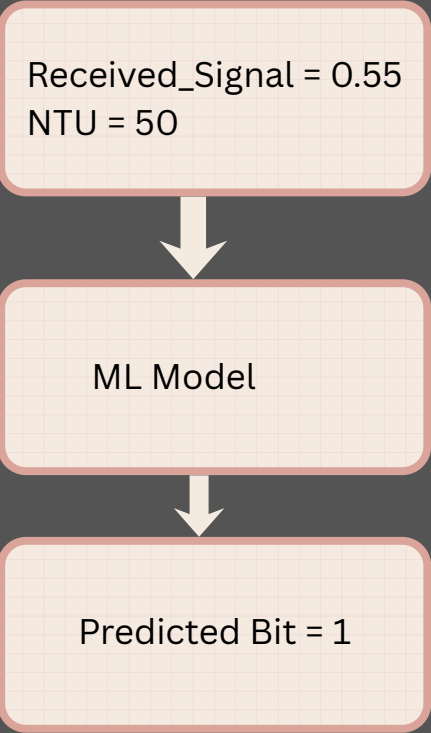
Prasanth M

Day 1 - Problem Understanding & System Definition

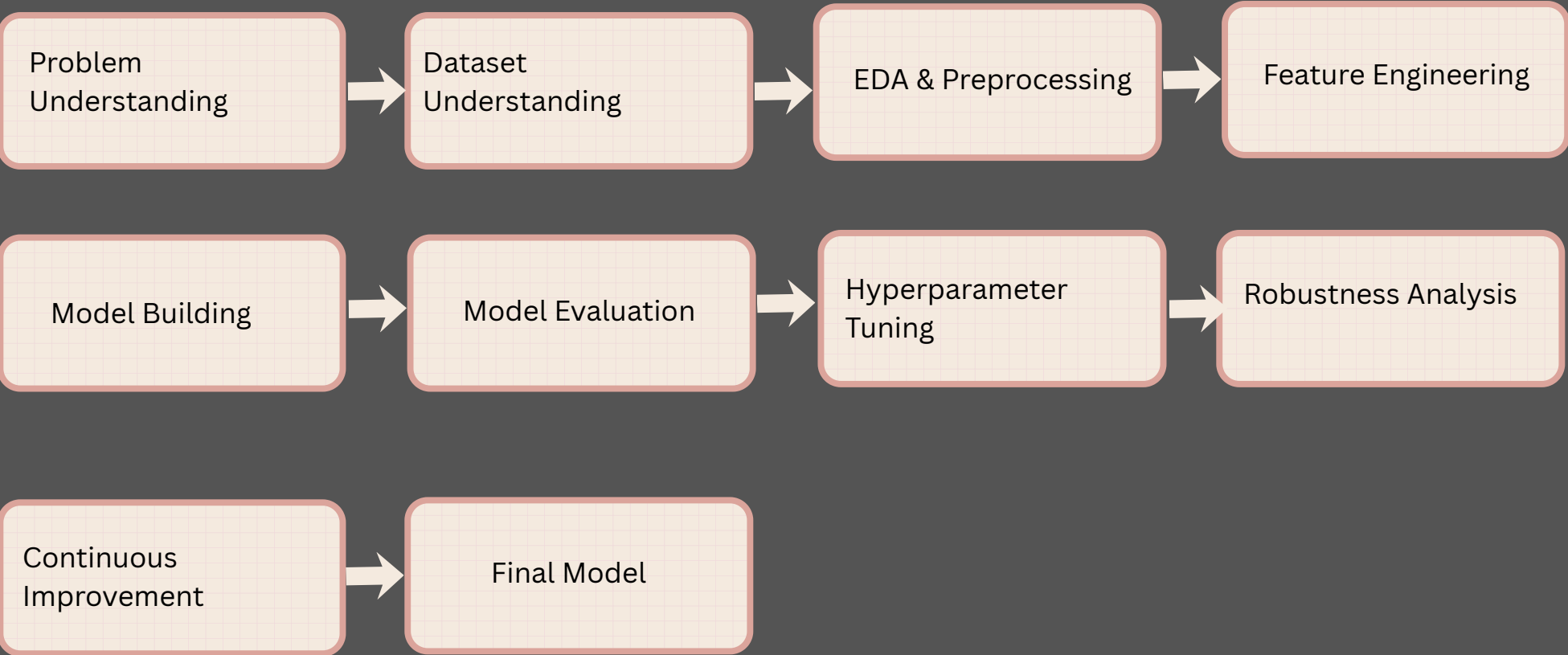
Problem Statement :

- Information is transmitted as binary bits (0 and 1) using optical signals underwater.
- Turbidity (NTU) causes absorption and scattering of light in water.
- This changes the received optical signal amplitude.
- As a result, fixed-threshold detection becomes unreliable and may cause bit errors.
- The dataset contains received analog signals, NTU values, and transmitted bit labels (0/1).
- The goal is to develop supervised ML models that accurately recover transmitted bits under different turbidity conditions.

Received_Sign	NTU	Transmitted_B
0.12	5	0
0.18	5	0
0.85	5	1
0.92	5	1
0.25	20	0
0.35	20	0
0.72	20	1
0.81	20	1
0.4	50	0
0.55	50	1
0.63	50	1



Pipeline | Workflow :



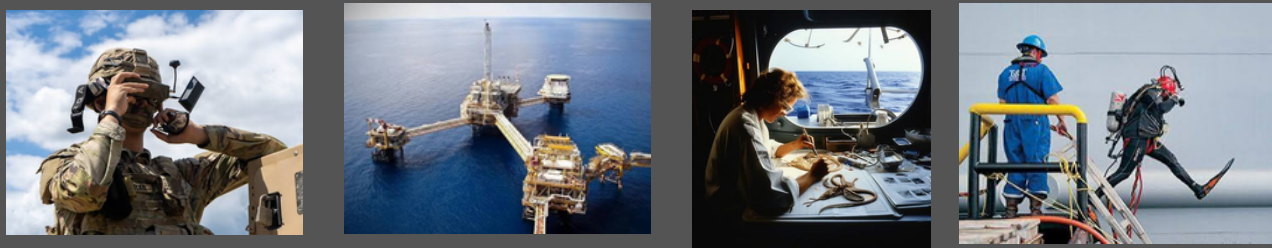
Day 2 - Literature Review - Domain + ML

UnderWater Communication :

Underwater communication is the transmission of information between underwater devices using sound, radio waves, or light.

Why is it Needed :

- Defense & Naval Operations: Communication between submarines, ships, and divers.
- Marine Research: Ocean monitoring, climate studies, and marine life observation.
- Offshore Industry: Pipeline inspection and underwater equipment monitoring.
- Rescue Operations: Diver-to-diver and diver-to-surface communication.

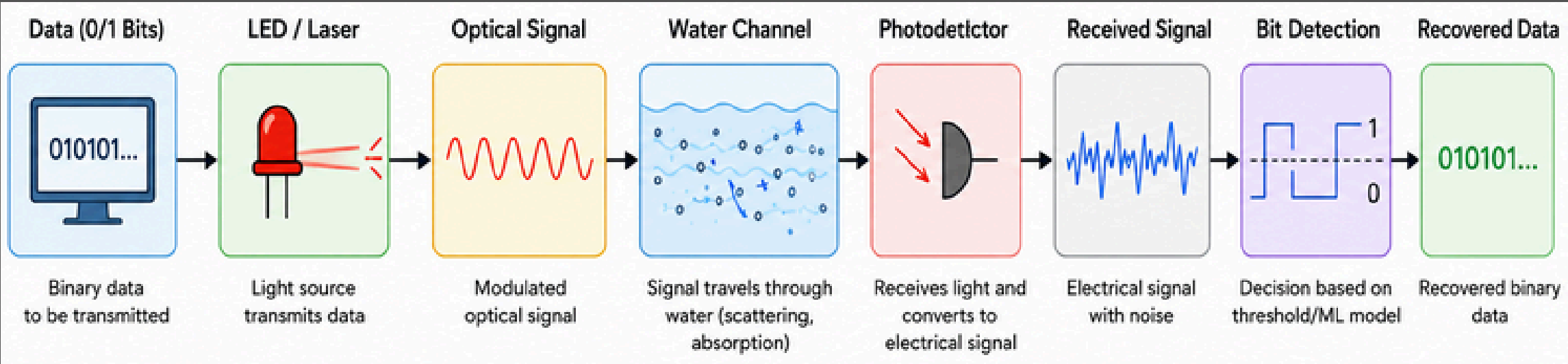


Communication Techniques :

Technique	Range	Bandwidth
Acoustic	Long (km)	Low (kbps)
Radio Frequency	Short	Moderate
Optical (UWOC)	Short	High (Gbps)



UWOC Working



What is Fixed Threshold ?

A Fixed Threshold is a simple method used to detect transmitted bits based on the received signal strength.

Received Signal > Threshold → Bit 1
Received Signal ≤ Threshold → Bit 0

Received Signal	Predicted Bit
0.8	1
0.7	1
0.2	0
0.3	0

Threshold = 0.5

How Turbidity Affects It ?

Turbidity (cloudiness of water) causes absorption and scattering of light.

As turbidity increases:

- Received signal becomes weaker.
- Signal values of bit 0 and 1 start overlapping.
- Fixed threshold can no longer separate them correctly.

Threshold = 0.5

Actual Bit	Received Signal	Predicted
0	0.55	1 ❌
1	0.48	0 ❌

Drawback :

- Same threshold is used for all water conditions.
- Does not adapt to different NTU levels.
- Leads to higher Bit Error Rate (BER).

Machine Learning :

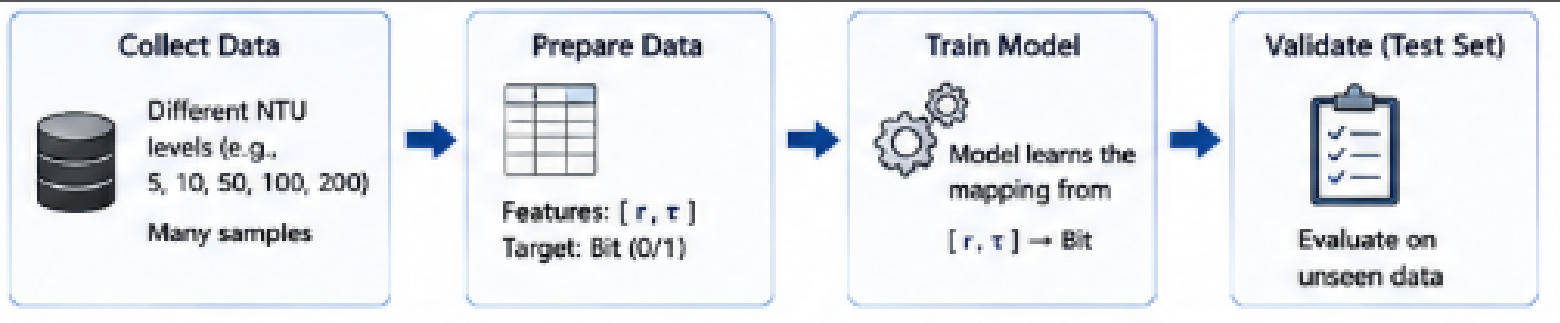
The idea is to let the ML model learn the relationship between the received optical signal, turbidity (NTU), and the actual transmitted bit instead of using one fixed threshold

ML Type : Supervised Machine Learning (Binary Classification)

- Goal: Classify the transmitted bit as 0 or 1.
- Input: Received Signal + NTU
- Output: Predicted Bit (0/1)

Working :

- Multiple Inputs: Uses received signal and NTU/turbidity.
- Learns Patterns: Learns from labelled 0 and 1 bits.
- Handles Turbidity: Learns signal patterns across different NTU levels.
- Reduces Errors: Improves bit detection and reduces BER.
- Compares Models: Compares ML models with fixed threshold using accuracy, F1, BER, and inference time.



Reviewed Methods :

Method	Approach	Advantages	Limitations
Fixed Threshold	Rule-based	Simple, Fast	Fails under varying turbidity
Logistic Regression	Linear Classifier	Interpretable	Limited for complex patterns
SVM	Margin-based Classification	Good generalization	Higher computation
Random Forest	Ensemble Trees	Handles non-linearity	Larger model size
MLP (Optional)	Neural Network	Learns complex patterns	Requires more training data

Evaluation Metrics :

- Accuracy: Overall correct predictions.
- Precision: Correctly predicted positive bits.
- Recall: Detected actual positive bits.
- F1-Score: Balance of precision and recall.
- Confusion Matrix: Shows classification errors.
- BER: Percentage of incorrectly detected bits.
- Inference Time: Time taken for prediction.

Success Criteria :

- High Accuracy: Correctly classify most bits.
- Low BER: Minimize bit detection errors.
- Robust Across NTU: Consistent performance at different turbidity levels.
- Fast Inference: Quick prediction with low computational cost.

Day 3 - Mentor Review

1. Introduction Problem Statement : Underwater Optical Wireless Communication (UWOC)

In UWOC, information is transmitted as binary bits (0 and 1) using optical signals. As light travels through water, turbidity (NTU) causes absorption and scattering, which changes the received optical signal amplitude. These variations make fixed-threshold detection unreliable and can lead to bit errors. To overcome this challenge, supervised machine learning models are used to classify transmitted bits based on received signal values and turbidity information, enabling more robust communication under varying underwater conditions.

2. Solution for this : ML

- Domain: Underwater Optical Wireless Communication (UWOC)
- Problem Type: Binary Classification
- Learning Type: Supervised Machine Learning
- Target Classes: Bit 0 and Bit 1

3. Objectives :

- Understand the UWOC problem and binary bit classification (0/1).
- Analyze received signal and NTU/turbidity data.
- Build a preprocessing and exploratory data analysis (EDA) pipeline.
- Implement and compare Fixed Threshold, Logistic Regression, SVM, and Random Forest.
- Evaluate using Accuracy, Precision, Recall, F1-Score, BER, and Inference Time.
- Test robustness across different NTU levels and identify limitations.



4. Tasks for 4 weeks :

- Problem Understanding & Literature Review
- Dataset Understanding & Quality Assessment
- Data Preprocessing & Exploratory Analysis
- Feature / Representation Preparation
- Baseline & Main Model Development
- Tuning & Project-Specific Analysis
- Validation, Error & Failure Analysis
- Documentation & Presentation

Day 4 - Dataset Design & Quality Expectations

Expected Dataset Structure

Column	Description
Received Signal	Analog signal value received at the receiver
NTU	Turbidity level of the water
Transmitted Bit	Actual bit label (0 or 1)
Metadata (Optional)	Sample ID, timestamp, place,

Possible Dataset Risks :

- Missing Data → Missing signal or NTU values.
- Wrong Labels → Incorrect bit labels (0/1).
- Class Imbalance → More samples of one bit than the other.

Dataset Risk Checklist :

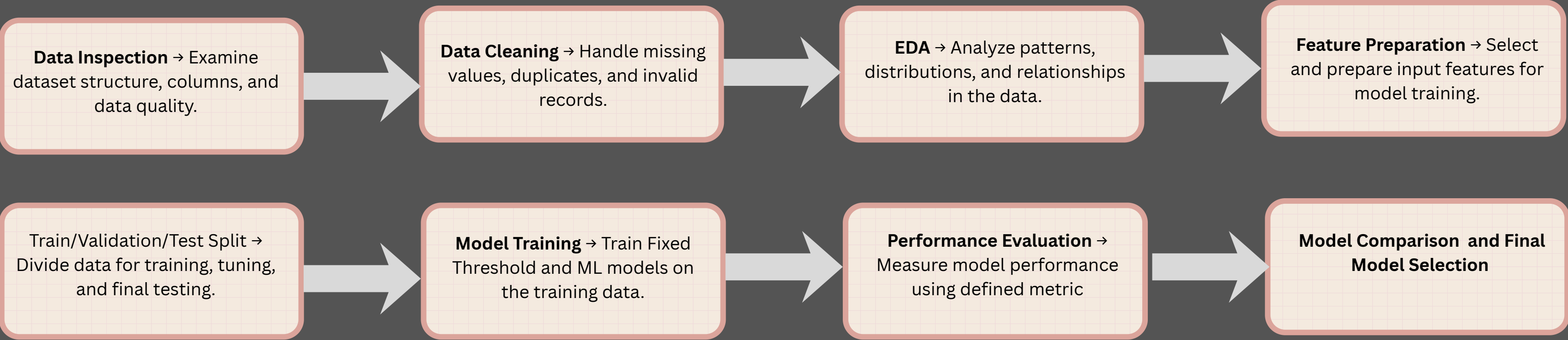
- Check missing values
- Check duplicate records
- Verify bit labels (0/1)
- Check class balance
- Check NTU coverage → Some turbidity levels may have very few samples.
- Check data leakage → Similar samples appearing in both train and test sets.
- Validate train-test split :

Train Set (70%) → Model learning.
Validation Set (15%) → Model tuning.
Test Set (15%) → Final evaluation.



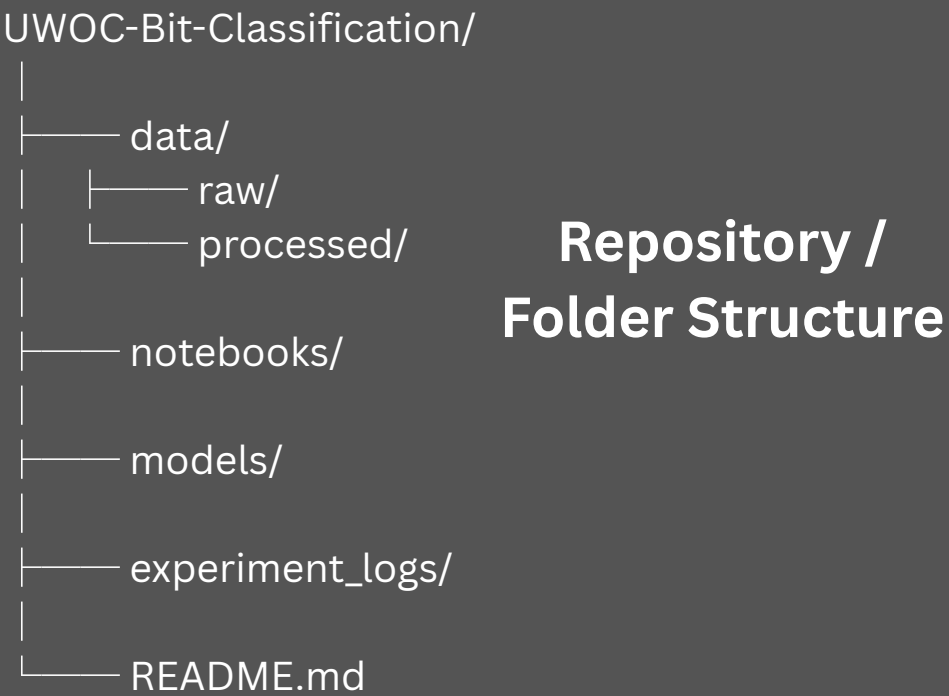
Day 5 - Pipeline & Experiment Planning

Pipeline :



Experiment Log Table

Model	Accuracy	Precision	Recall	F1-Score	BER	Inference Time
Fixed Threshold	-	-	-	-	-	-
Logistic Regression	-	-	-	-	-	-
SVM	-	-	-	-	-	-
Random Forest	-	-	-	-	-	-
MLP	-	-	-	-	-	-



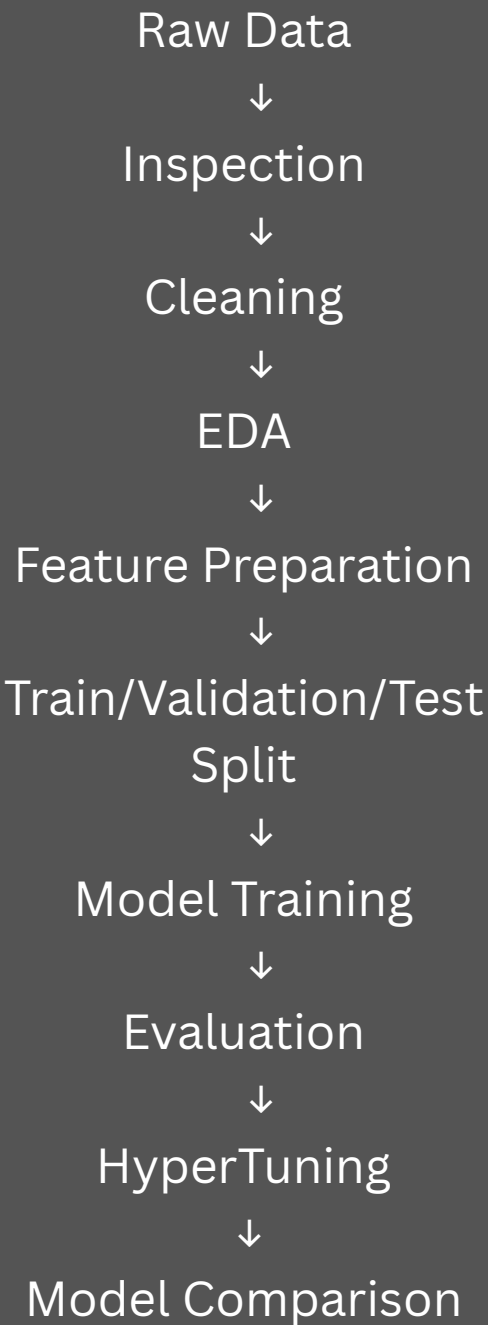
Day 6 -Pre-Dataset Readiness + Mentor Review

Progress Completed :

- ✓ Understood the UWOC problem statement and project objectives.
- ✓ Questions Discussed with MentorStudied the impact of turbidity (NTU) on optical signal transmission.
- ✓ Defined inputs, outputs, success metrics, and project scope.
- ✓ Designed the complete ML pipeline and experiment plan.
- ✓ Prepared dataset schema, risk checklist, and repository structure.

Mentor Discussion

1. Pipeline Discussion



2. Grid Search vs Random Search

Grid Search

- Tests every possible parameter combination.
- More exhaustive but computationally expensive.

Random Search

- Tests randomly selected parameter combinations.
- Faster and often finds good solutions with less computation.

3. Why RF Communication Does Not Work Well Underwater?

- Water strongly absorbs high-frequency radio waves.
- Signal attenuation is very high underwater.
- Communication range becomes extremely limited.

4. How Does Turbidity Affect Communication?

- Suspended particles absorb and scatter light.
- Received optical signal becomes weaker and noisier.
- Bit 0 and Bit 1 signal values may overlap.
- Fixed-threshold detection becomes unreliable.

Next Steps :

- Receive and inspect the dataset.
- Perform data quality audit.
- Apply approved preprocessing strategy.
- Begin baseline and ML model implementation.



Day 7 - Dataset Receipt & Initial Inspection

Dataset Description :

- Dataset Shape : 399 Rows and 6 columns are available
- Column Description :
 - binary bit R → Received bit (after detection)
 - Binary bit S → Sent/Transmitted bit (Target Label)
 - Analog value → ADC reading (Feature)
 - Analog Voltage → Converted ADC voltage (Feature)
 - NTU → Turbidity level (Feature)
 - S No → Just index (drop it)
- Dataset size : 12 Kb

S No	binary bit	Binary bit	Analog value	Analog Voltage	NTU
1	1	1	125	0.610351563	1-2NTU
2	0	0	101	0.493164063	1-2NTU
3	1	1	123	0.600585938	1-2NTU
4	0	0	101	0.493164063	1-2NTU
5	0	0	95	0.463867188	1-2NTU
6	0	0	99	0.483398438	1-2NTU
7	1	1	142	0.693359375	1-2NTU
8	0	0	97	0.473632813	1-2NTU
9	0	0	98	0.478515625	1-2NTU
10	0	0	98	0.478515625	1-2NTU

Input Features:

- Analog value
- Analog Voltage
- NTU

Target:

- Binary bit S

Unncessary Feature :

- S.no - No use
- Binary bit R we need correct bit which is binary bit s



Day 8 - Data Quality & Class Audit



Data Quality Table

Metric	Value
Total Rows	399
Total Columns	6
Missing Values	0
Duplicate Rows	0
Invalid Bit Labels	0
Class 0 Samples	242
Class 1 Samples	157
Class 0 Percentage	60.65%
Class 1 Percentage	39.35%
Class Balance Status	Moderately Imbalanced

Obseravtion :

- Dataset contains 399 samples and 6 features.
- No missing values were found.
- No duplicate records were detected.
- No invalid bit labels were present.
- Target classes are moderately imbalanced (Class 0: 60.65%, Class 1: 39.35%).
- Overall, the dataset is clean and ready for preprocessing and model training.

Cleaning Plan :

- 1.Drop the S No column as it does not contribute to prediction.
- 2.Verify and standardize NTU labels if multiple formats exist.
- 3.Ensure Analog Value and Analog Voltage are stored as numeric data types.
- 4.Check for class imbalance and use stratified train-validation-test splitting.
- 5.Preserve the distribution of both Binary bit S classes and NTU levels across all splits.
- 6.Scale numerical features if required by models such as Logistic Regression or SVM.



Day 9 - Cleaning & Split Review

Data Quality Findings :

- Total samples: 399
- Total features: 6
- Missing values: 0
- Duplicate records: 0
- Invalid bit labels: 0
- Dataset is clean with no major quality issues detected.



Class Balance Findings :

Class	Count	Percentage
0	242	60.65%
1	157	39.35%

- Dataset shows a moderate class imbalance.
- Both classes are sufficiently represented for model training.

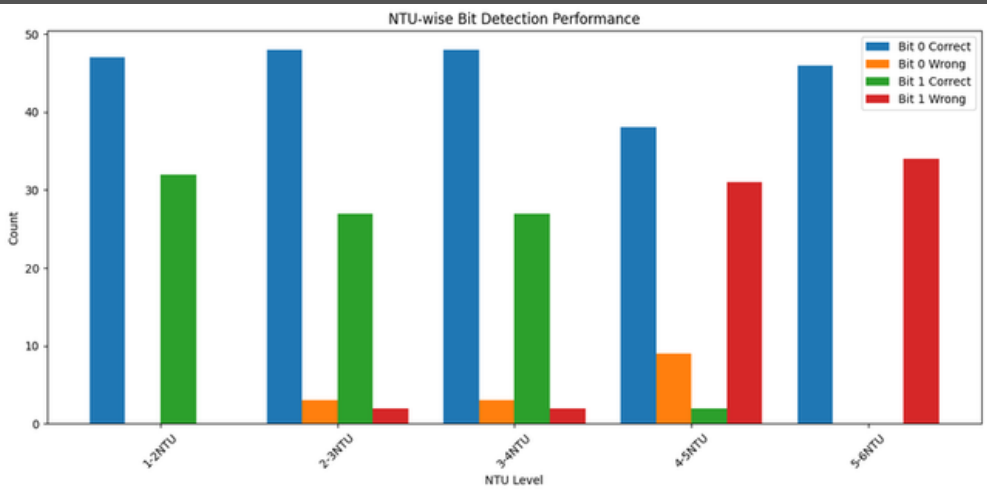
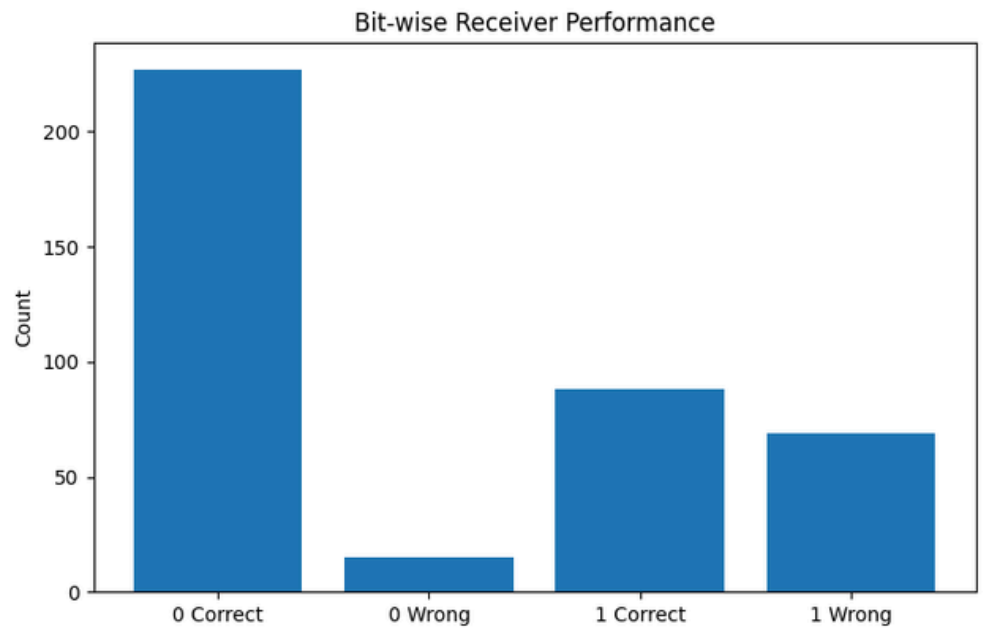
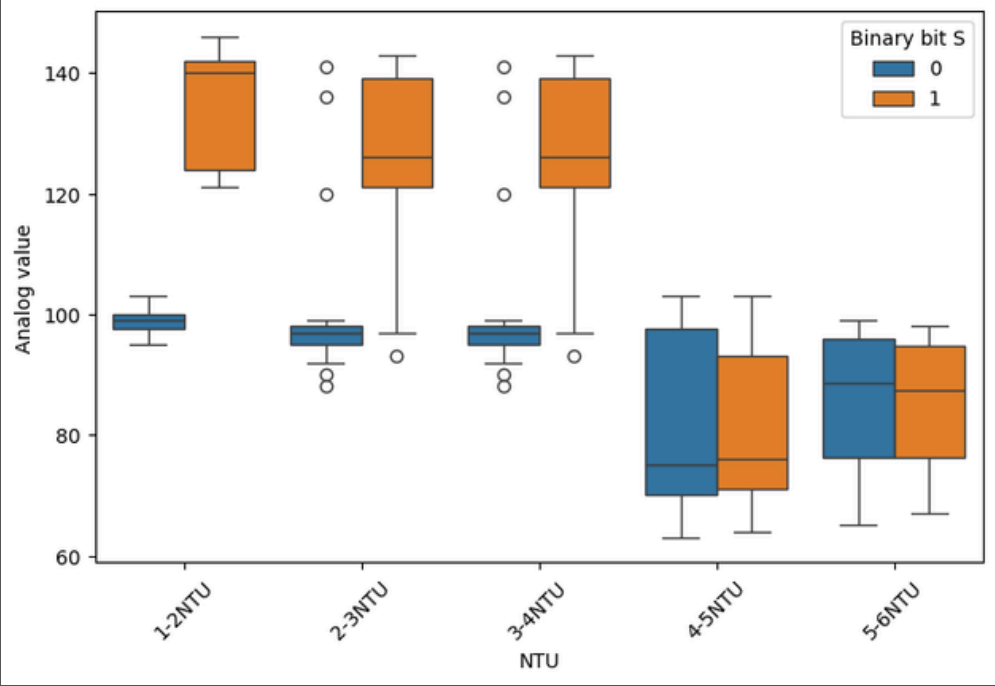
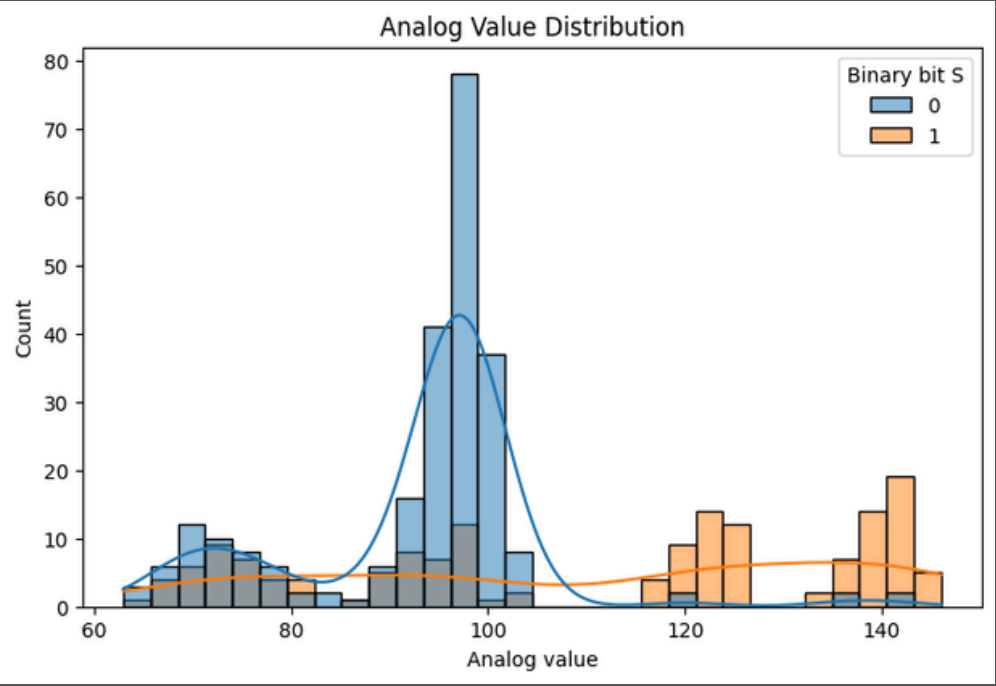
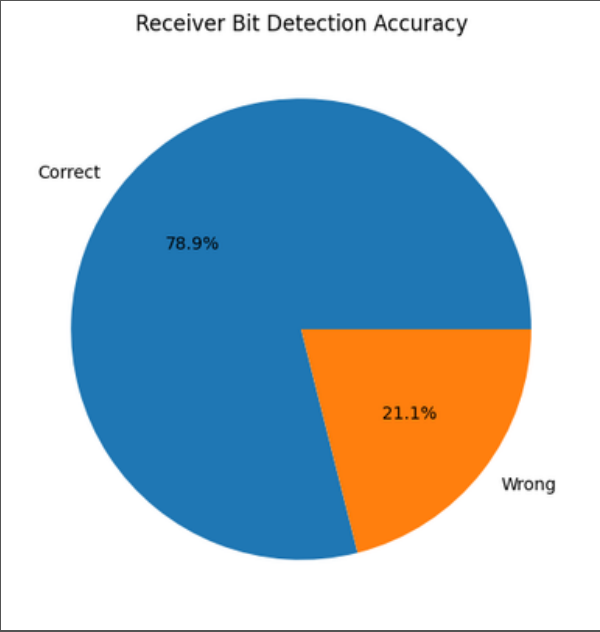
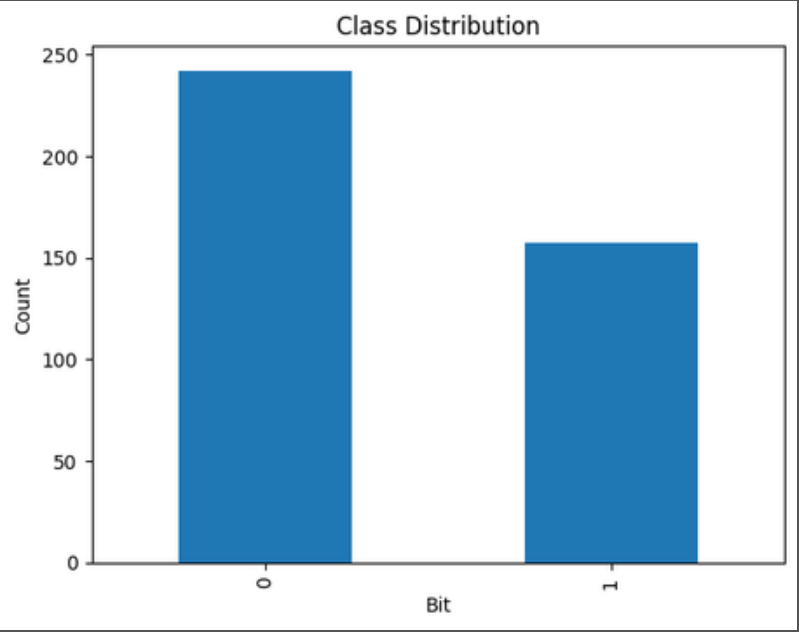
Proposed Split Strategy :

Dataset Split	Percentage
Training Set	70%
Validation Set	15%
Test Set	15%

Proposed Scaling Strategy :

- Drop S No column.
- Encode NTU if required.
- Apply StandardScaler for:
 - Analog Value
 - Analog Voltage
- Required for Logistic Regression, SVM, and MLP.
- Random Forest can be trained with or without scaling.

Day 10 - EDA & Class Separability

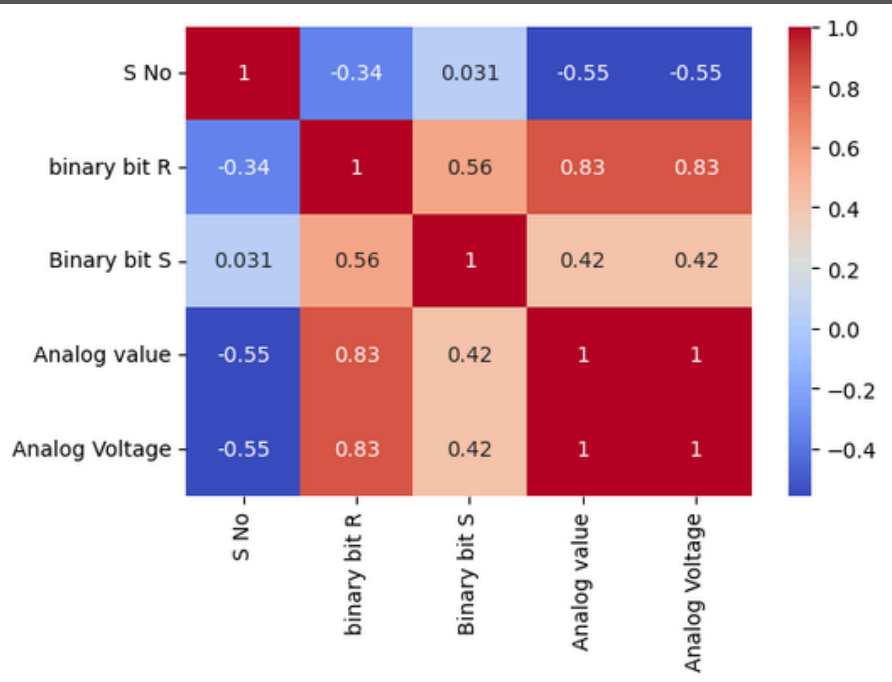


Observation :

- The receiver detects bits correctly 78.9% of the time.
- The test dataset has more Bit 0 samples than Bit 1 samples.
- The system detects Bit 0 much more accurately than Bit 1.
- Detection performance drops significantly as water turbidity increases above 4 NTU.
- High turbidity completely blocks Bit 1 detection, making all signals look like Bit 0.

Observation :

- In clearer water (1-4 NTU), Bit 1 produces much higher analog values than Bit 0, making the two bits easy to separate.
- In murkier water (4-6 NTU), the analog signal drops sharply, causing both bits to overlap and become indistinguishable.
- Overall, Bit 0 stays consistently centered near an analog value of 100, while Bit 1 weakens and scatters across lower values as turbidity rises.



Observation :

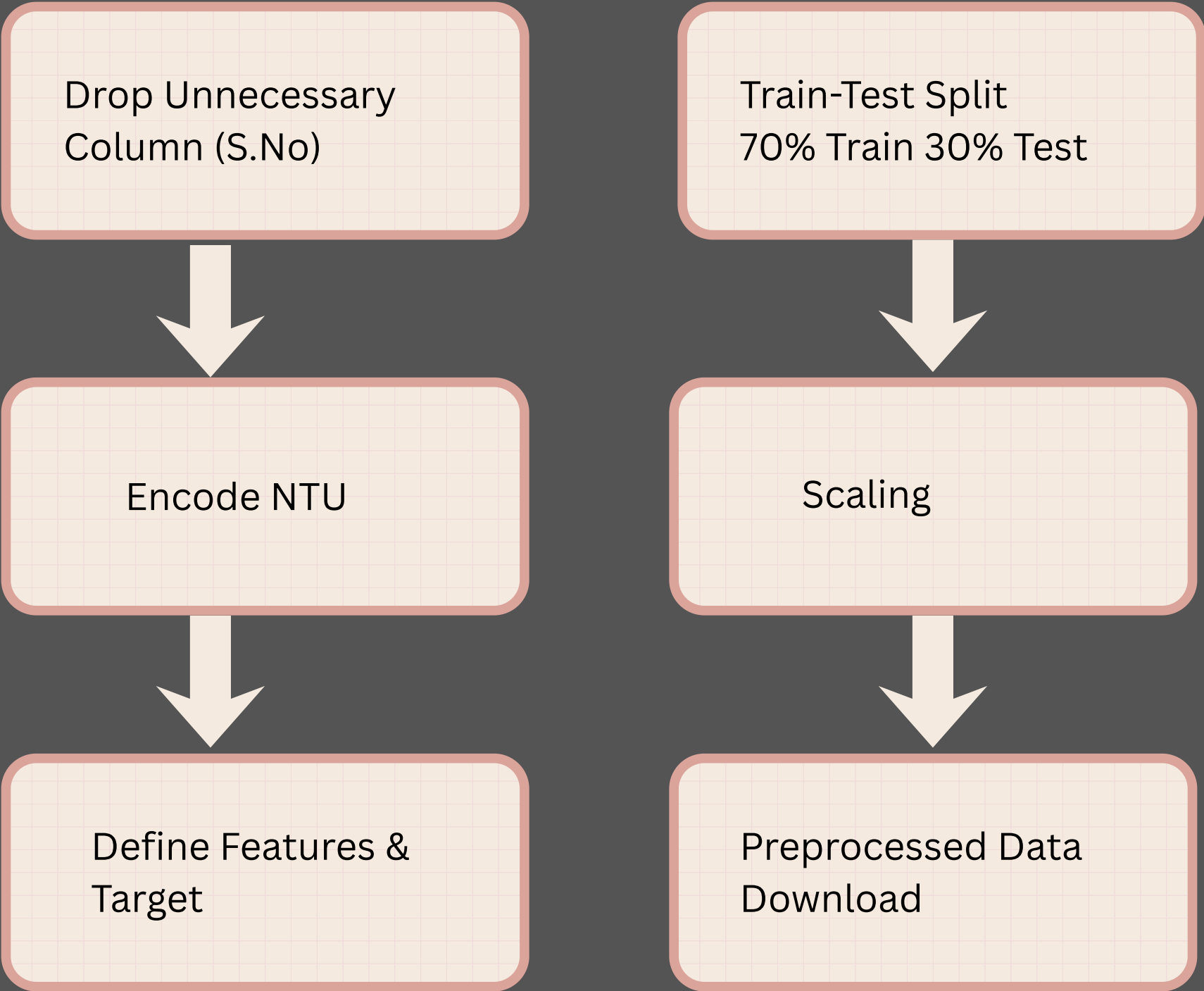
Binary Bit s has some moderate correlation ship with the analong voltage and value



Day 11 and 12 - Preprocessing and Review



Preprocessing Pipeline



PreProcessed Data link

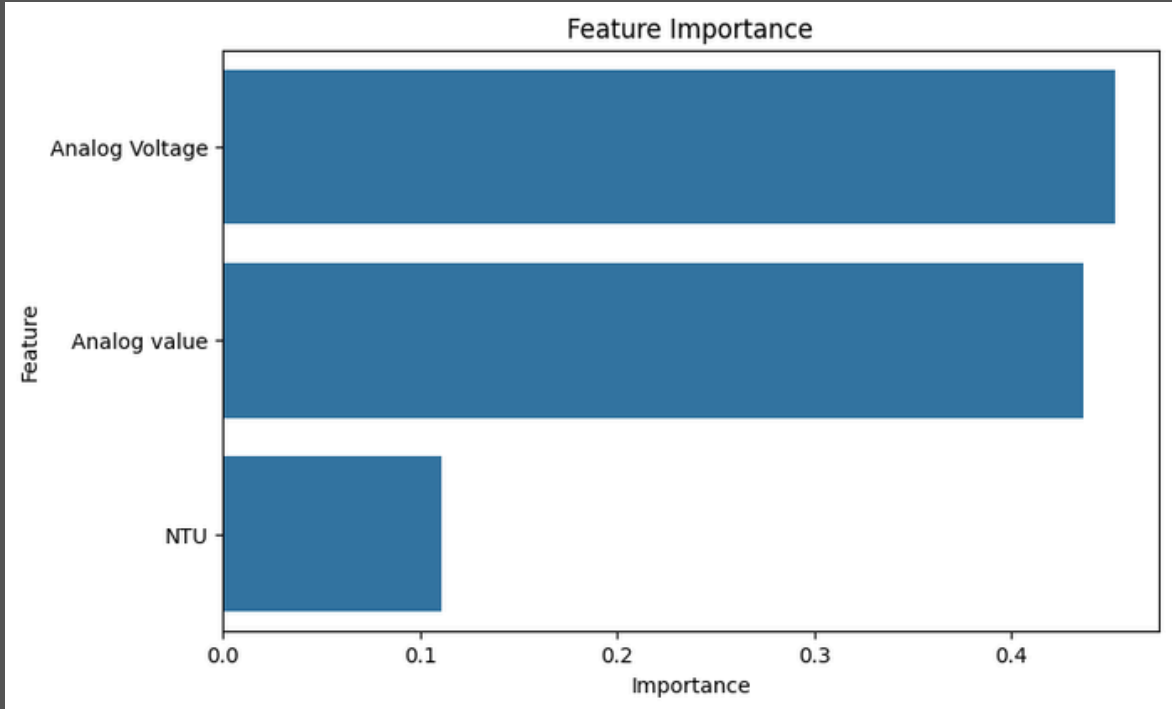
Preprocessed Data

	Analog value	Analog Voltage	NTU
0	-0.895865	-0.895865	1.509531
1	0.996565	0.996565	-1.360378
2	-0.750293	-0.750293	1.509531
3	-0.556198	-0.556198	1.509531
4	1.142136	1.142136	-1.360378

Preprocessing Summary :

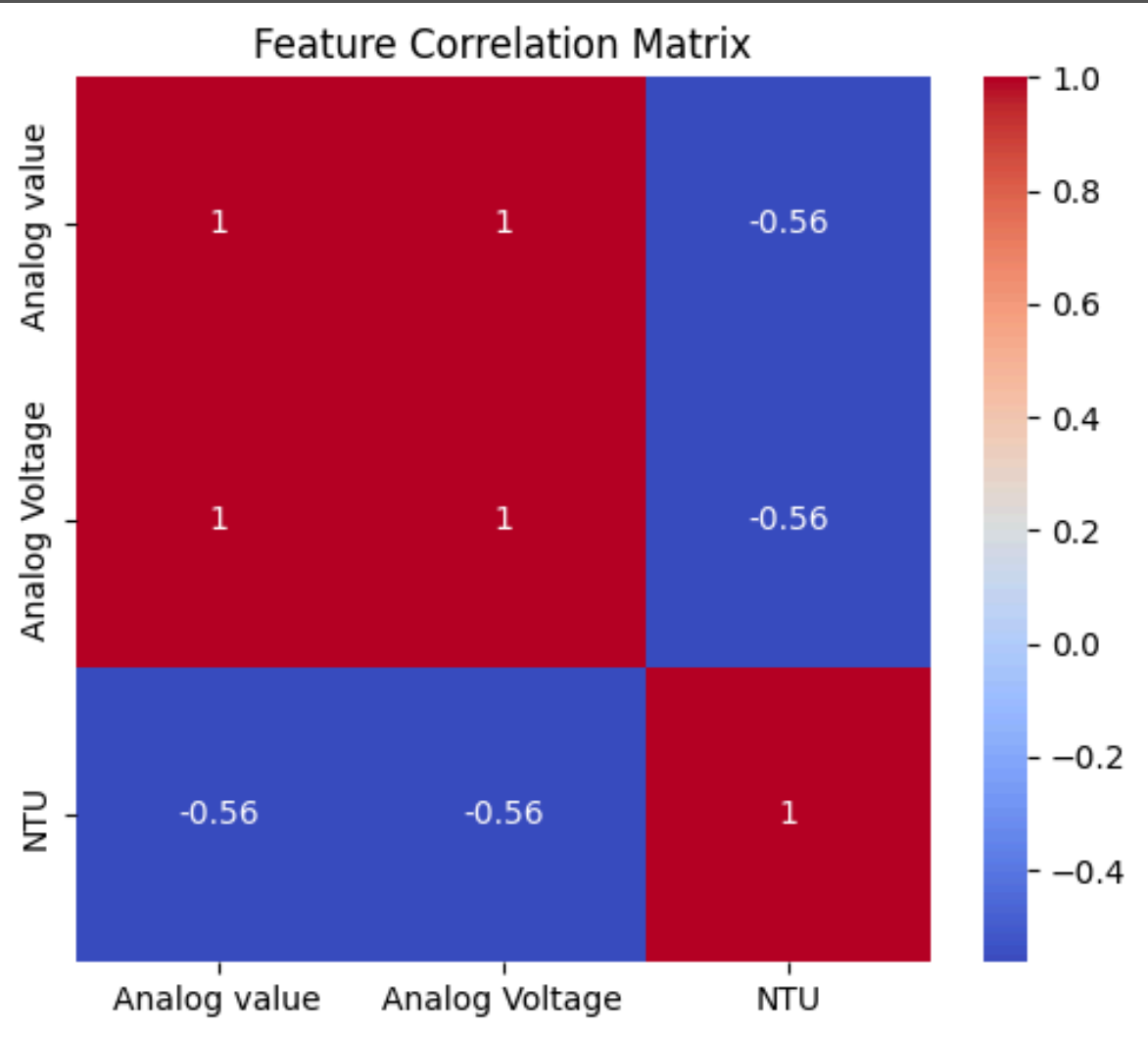
- 1.Removed unnecessary columns (e.g., S No) that do not contribute to prediction.
- 2.Checked data quality and confirmed no missing values, duplicates, or invalid labels.
- 3.Encoded NTU levels into numerical values for machine learning models.
- 4.Applied stratified train-test split to preserve class distribution across datasets.
- 5.Scaled input features (Analog Value, Analog Voltage, NTU) to improve model training and performance.

Day 13 - Feature Importance



Feature Importance Table

Rank	Feature	Importance
1	Analog Voltage	0.45
2	Analog Value	0.44
3	NTU	0.11



Feature Engineering Observations :

- Analog Value and Analog Voltage are perfectly correlated (1.00), indicating they contain nearly identical information.
- NTU has a moderate negative correlation (-0.56) with the signal features, showing that turbidity affects the received signal.
- Analog Voltage and Analog Value are the most important features for bit classification.
- NTU contributes useful environmental information but has lower predictive importance compared to signal features.
- Since Analog Voltage is directly derived from Analog Value, one of them may be removed in future experiments to reduce feature redundancy.

Day 14 - Peer Consolidation

My EDA Findings



No missing values, duplicate records, or invalid bit labels were found



The dataset shows a moderate class imbalance (60.65% class 0, 39.35% class 1).



Analog Value and Analog Voltage are the most important features for bit classification.



NTU has a noticeable impact on signal behavior and classification performance.



Analog Value and Analog Voltage are highly correlated, indicating feature redundancy.

Difficult Classes / Conditions:

Bit classification becomes more challenging at higher turbidity (NTU) levels due to increased signal degradation.

Peer Insight :

- Comparing performance across individual NTU levels may reveal how turbidity affects classification accuracy and BER.

Hypothesis :

- As NTU increases, the received optical signal becomes more distorted, causing fixed-threshold detection performance to decrease, while machine learning models can better adapt to changing signal patterns and maintain higher classification accuracy.



Day 18 - Model / Method Comparison

Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	BER
Logistic Regressi	0.742	0.667	0.681	0.674	0.258
SVM	0.758	0.846	0.468	0.603	0.242
Random Forest	0.708	0.63	0.617	0.624	0.292
MLP	0.75	0.717	0.596	0.651	0.25

Data Limitations

- Dataset size is relatively small (399 samples).
- Analog Value and Analog Voltage are highly correlated (redundant information).
- Limited number of features.
- Performance may vary under unseen NTU conditions.

Shortlisted Models for Deeper Experiments

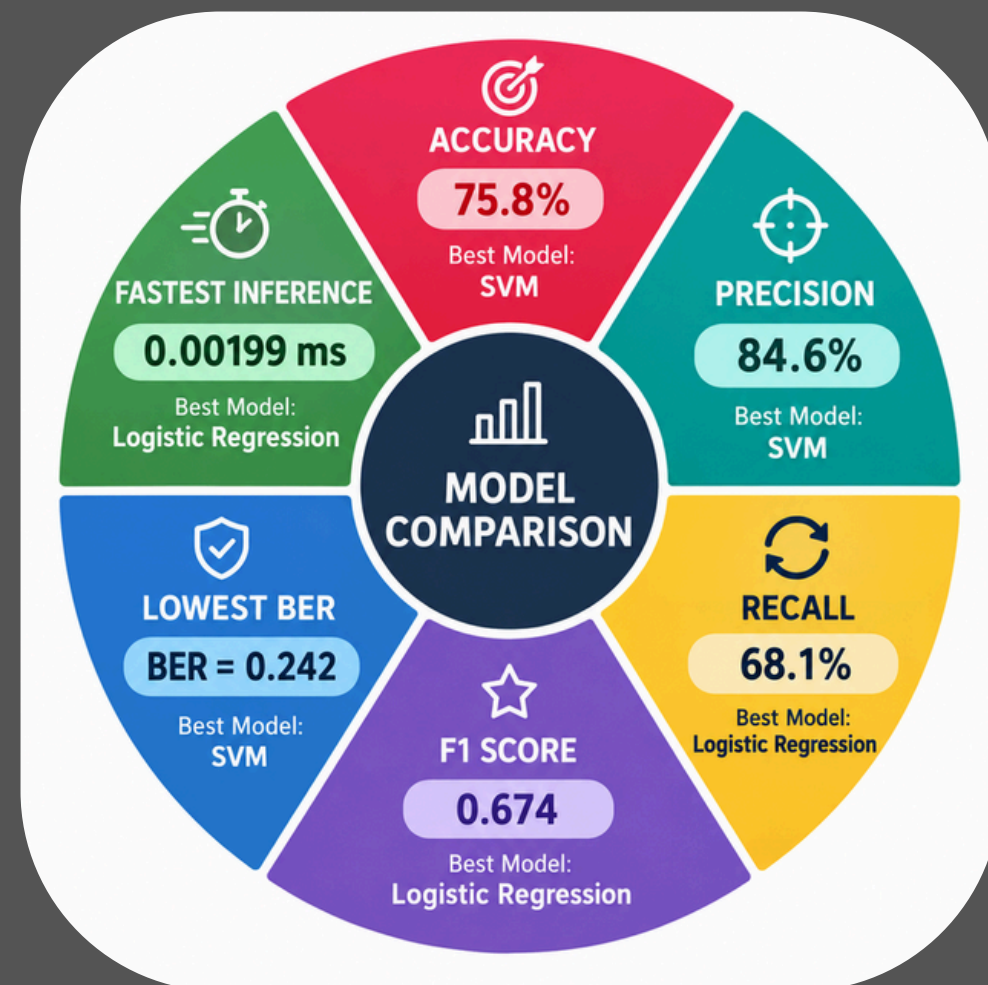
Candidate 1: SVM

- Highest Accuracy
- Highest Precision
- Lowest BER
- Strong classification performance



Candidate 2 : Logistic Regression

- Highest Recall
- Highest F1-Score
- Simple and interpretable
- Good baseline model



Day 19 - HyperParameter Tuning

Tuned candidates

Logistic Regression

- **C : Controls regularization.** [0.1,1,10]
Small C → More regularization.
→ Simpler model.
→ Less overfitting.
Large C → Less regularization.
→ More complex model.
- **Solver : optimization algorithm.**
lbfgs :
→ Modern optimizer.
→ Fast.
→ Works well for small/medium datasets.
liblinear :
→ Older optimizer.
→ Good for small datasets.

SVM

- **C : Controls regularization.** [0.1,1,10]
Small C → More regularization.
→ Simpler model.
→ Less overfitting.
Large C → Less regularization.
→ More complex model.
- **Kernel : Kernel determines the shape of the decision boundary.**
Linear Kernel → Straight line. Useful when data is linear
RBF Kernel → Curved boundary. Useful when data is nonlinear

{'C': 0.1, 'solver': 'lbfgs'}

**Best
parameter
after applying
Grid search cv**

{'C': 0.1, 'kernel': 'linear'}

Evaluation Metrics Using Tuned Models

Model	Accuracy	Precision	Recall	F1-Score	BER
Logistic Regression	0.742	0.69	0.617	0.651	0.258
SVM	0.758	0.846	0.468	0.603	0.242

Conclusion:

- Grid Search was performed on the shortlisted SVM and Logistic Regression models.
- The best SVM configuration was Linear Kernel with C=0.1.
- The best Logistic Regression configuration was C=0.1 with lbfgs solver.
- Cross-validation accuracy reached 82.81% for SVM and 79.25% for Logistic Regression.
- Final test-set performance remained similar to the baseline results, indicating that the original models were already close to optimal for the current dataset.
- SVM remained the strongest candidate and Logistic Regression remained the second-best candidate for further robustness analysis.

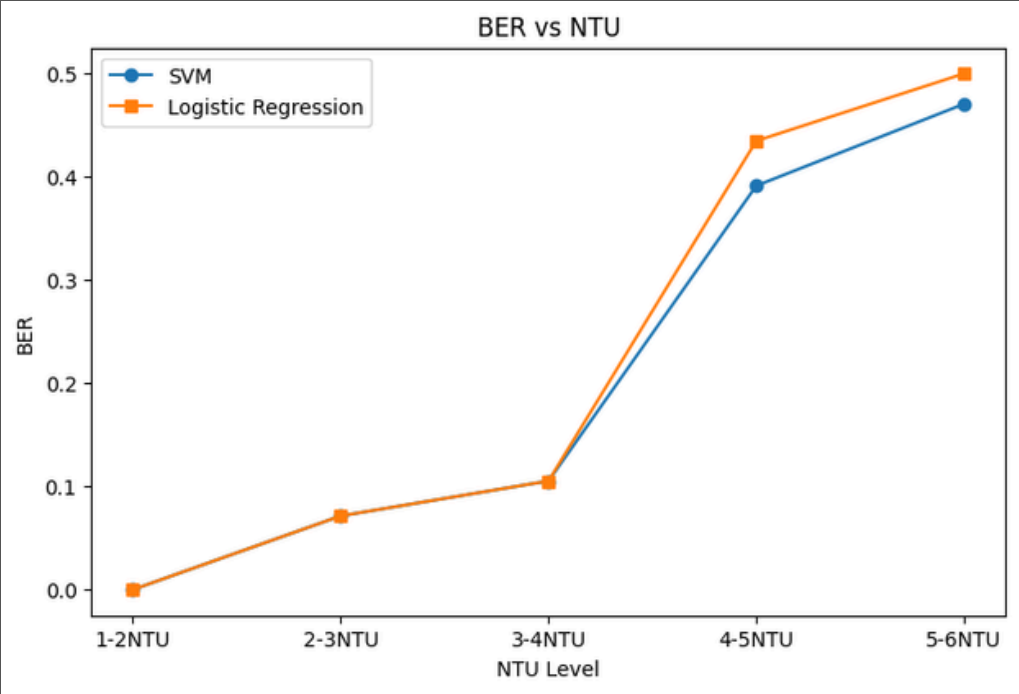
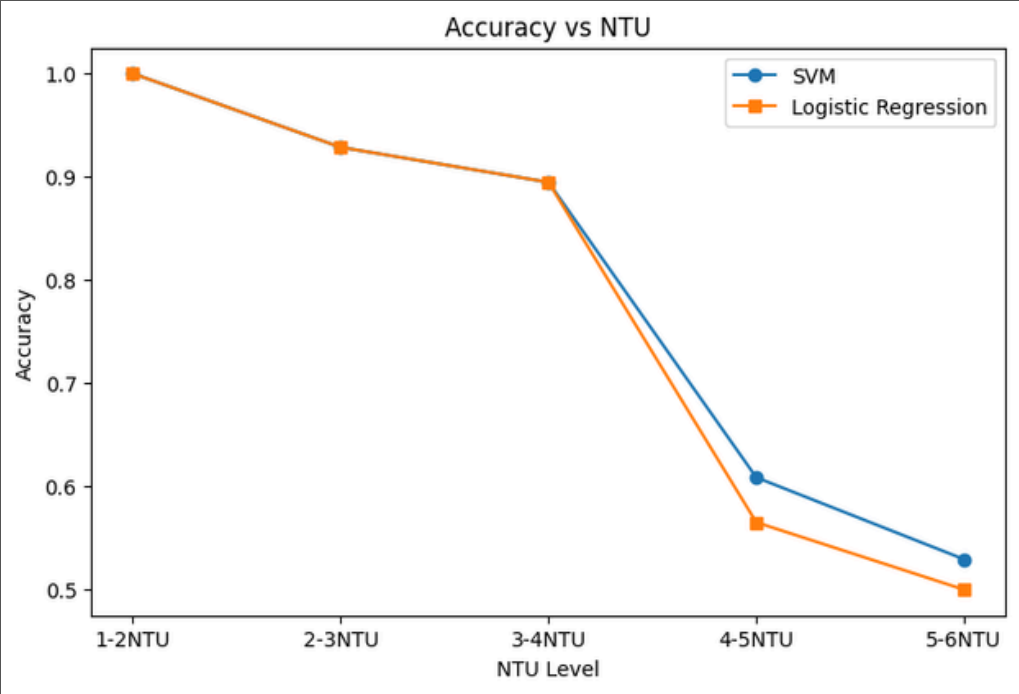
Day 20 - Robustness Study

SVM NTU-wise Results :

	NTU	Accuracy	Precision	Recall	F1	BER
0	1-2NTU	1.000000	1.000000	1.0	1.000	0.000000
1	2-3NTU	0.928571	0.818182	1.0	0.900	0.071429
2	3-4NTU	0.894737	0.777778	1.0	0.875	0.105263
3	4-5NTU	0.608696	0.000000	0.0	0.000	0.391304
4	5-6NTU	0.529412	0.000000	0.0	0.000	0.470588

Logistic Regression NTU-wise Results :

	NTU	Accuracy	Precision	Recall	F1	BER
0	1-2NTU	1.000000	1.000000	1.0	1.000000	0.000000
1	2-3NTU	0.928571	0.818182	1.0	0.900000	0.071429
2	3-4NTU	0.894737	0.777778	1.0	0.875000	0.105263
3	4-5NTU	0.565217	0.000000	0.0	0.000000	0.434783
4	5-6NTU	0.500000	0.470588	0.5	0.484848	0.500000



Research Insight :

Effect of Turbidity (NTU)

- Low NTU conditions (1-4 NTU) provide clear signal separation between transmitted bits.
- As NTU increases, scattering and absorption distort the received optical signal.
- Signal overlap between bit 0 and bit 1 increases.
- Classification becomes more difficult at higher turbidity levels.
- BER increases while Accuracy and F1-score decrease.



Day 21 - Best Candidate Review : SVM

Baseline vs Tuned Results

Model	Version	Accuracy	Precision	Recall	F1	BER
Logistic Regression	Baseline	74.17%	66.95%	61.70%	65.17%	0.258
Logistic Regression	Tuned (C=0.1, lbfgs)	74.17%	66.95%	61.70%	65.17%	0.258
SVM	Baseline	75.83%	84.62%	46.81%	60.27%	0.242
SVM	Tuned (C=0.1, Linear)	75.83%	84.62%	46.81%	60.27%	0.242

Observation :

- Hyperparameter tuning did not significantly improve test-set performance.
- Baseline models were already close to optimal.
- SVM remains the strongest overall model.
- Logistic Regression remains a strong secondary candidate.



Representative Successes :



Success Case 1

- Low NTU condition.
- Clear separation between bit 0 and bit 1.
- Both models correctly classified all bits.

Success Case 2

- Medium NTU condition.
- Small signal variations.
- Models maintained high accuracy (>89%).

Representative Failures :



Failure Case 1

- High NTU condition.
- Significant signal attenuation.
- Bit 0 and bit 1 values overlap.

Failure Case 2

- Samples near decision boundary.
- Model predicts incorrect bit.

Reson for Failure

- Small dataset size (399 samples).
- Limited feature set (Analog Value, Analog Voltage, NTU).
- Performance degrades under high turbidity conditions.
- No real-time underwater testing available.

Day 22 - Final Model / Pipeline Freeze

Final Model Selection : SVM

Reason for Selection

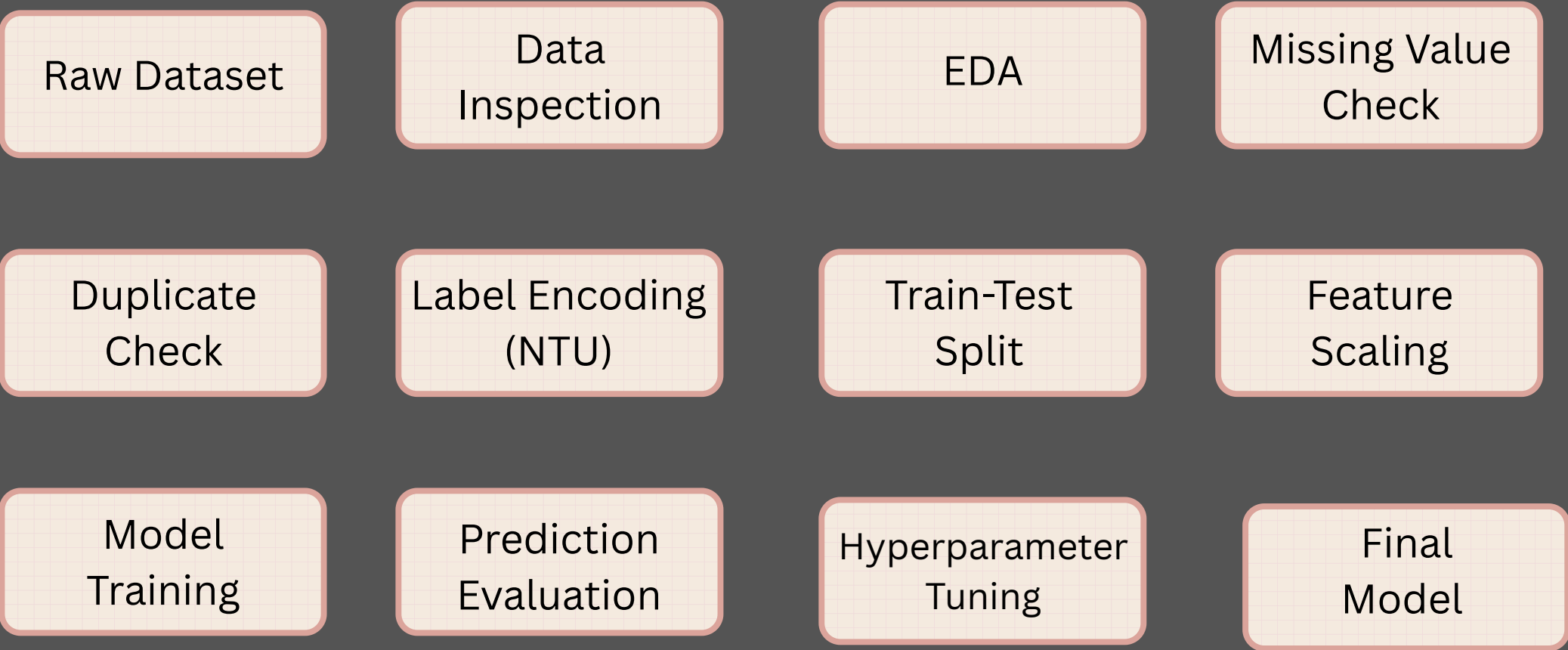
- Highest Accuracy (75.83%)
- Highest Precision (84.62%)
- Lowest BER (0.242)
- Better robustness at higher NTU levels

Frozen Hyperparameters :

SVM → C=0.1, Kernel=Linear



Frozen Preprocessing Pipeline



FINAL

Day 23 - Final Validation

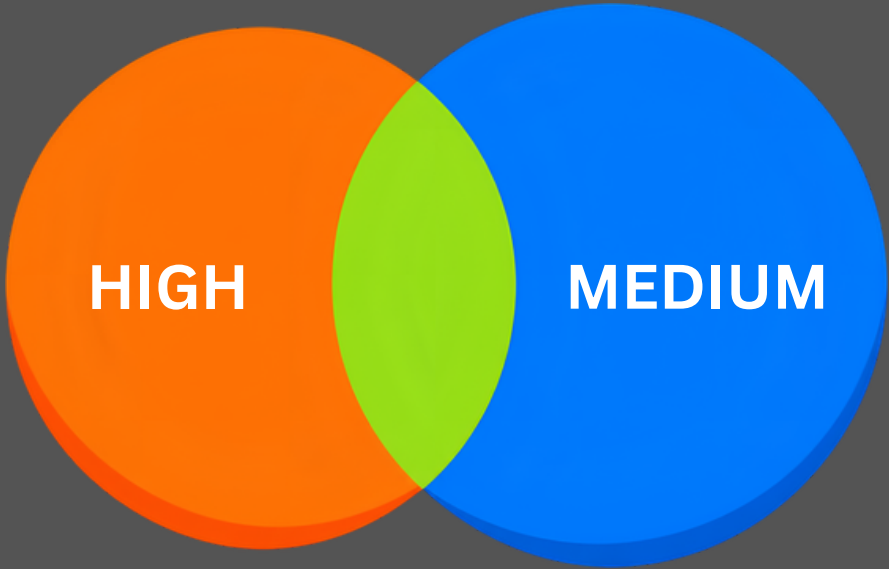
Final Validation Results

Metric	SVM
Accuracy	75.83%
Precision	84.62%
Recall	46.81%
F1 Score	60.27%
BER	0.242
Inference Time	Very Low

Runtime Analysis :

- Higher classification performance
- Better BER performance
- Slightly slower than Logistic Regression

Accuracy–Complexity Trade-off



SVM provides best classification performance,Also some complexity

Final Validation Summary :

- Final validation performed using held-out test data.
- No data leakage observed.
- Model performance consistent with cross-validation findings.
- SVM approved as final UWOC bit classifier.

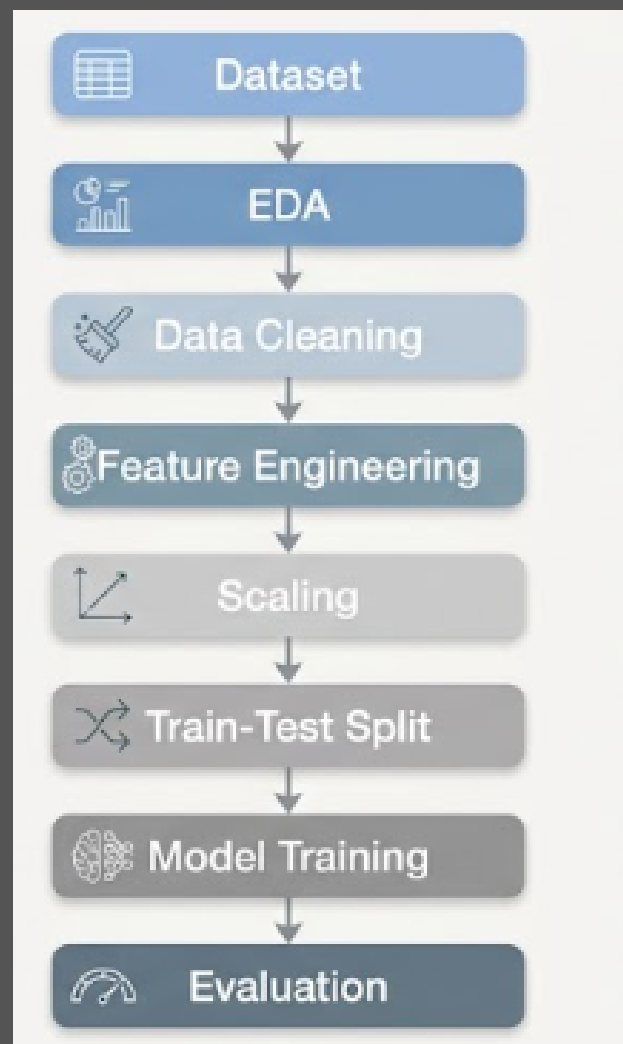


Day 24 - Final Technical Review - Summary

Problem Statement :

In Underwater Optical Communication (UWOC), turbidity (NTU) causes attenuation and scattering of optical signals. This changes the received signal strength, making fixed-threshold detection unreliable. The objective is to use supervised machine learning to classify transmitted bits (0 or 1) accurately under varying turbidity conditions.

Methodology



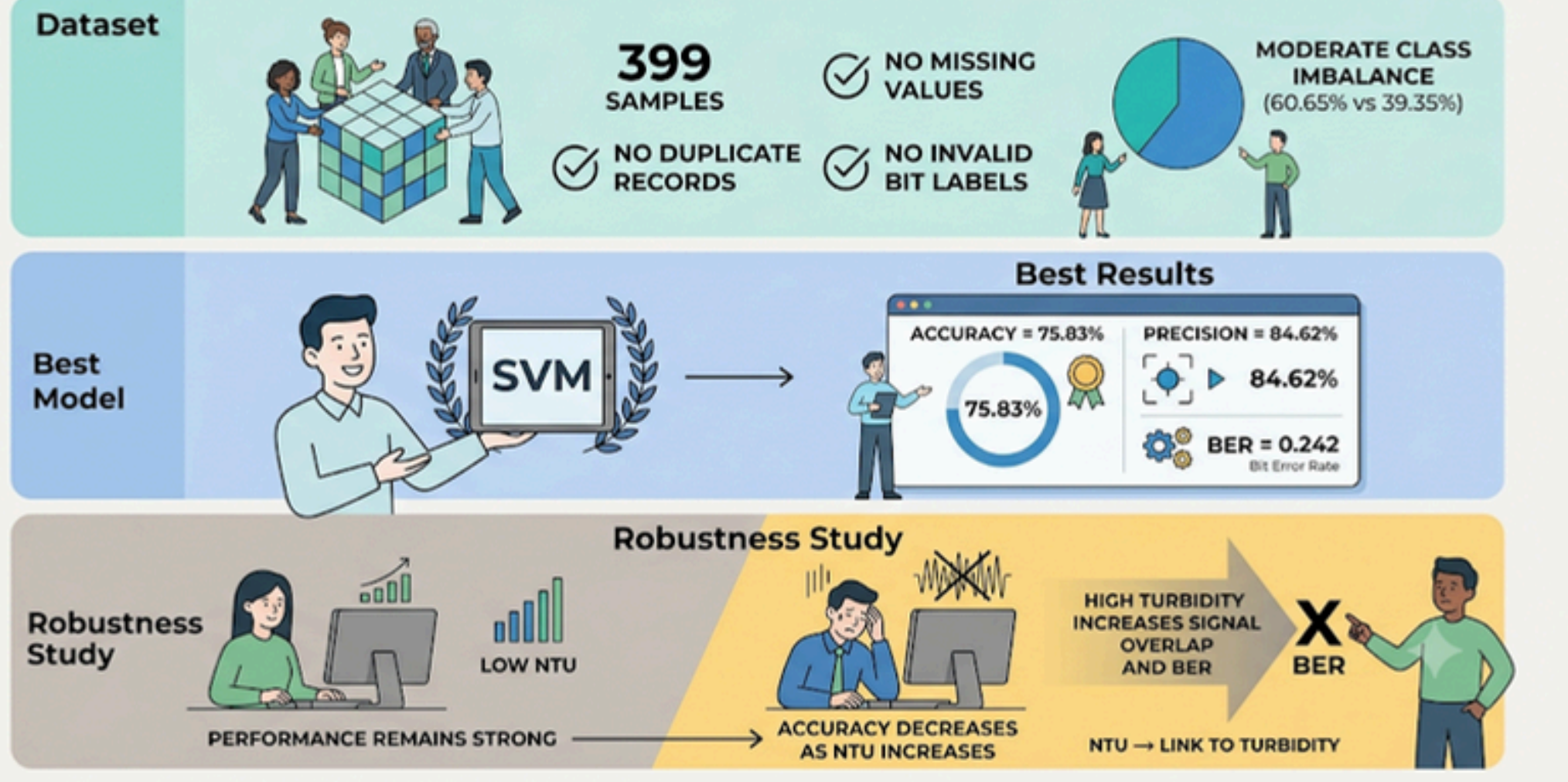
Models Evaluated:

- Fixed Threshold
- Logistic Regression
- SVM
- Random Forest
- MLP

Limitations :

- Small dataset size
- Moderate class imbalance
- Limited feature set
- Performance degradation at high NTU levels
- No real-world deployment testing

KEY FINDINGS



Future Work

- Collect larger UWOC datasets.
- Test advanced models (XGBoost, LightGBM, Deep Learning).
- Incorporate additional environmental features.
- Evaluate real-time deployment on embedded hardware.
- Improve robustness under high turbidity conditions.

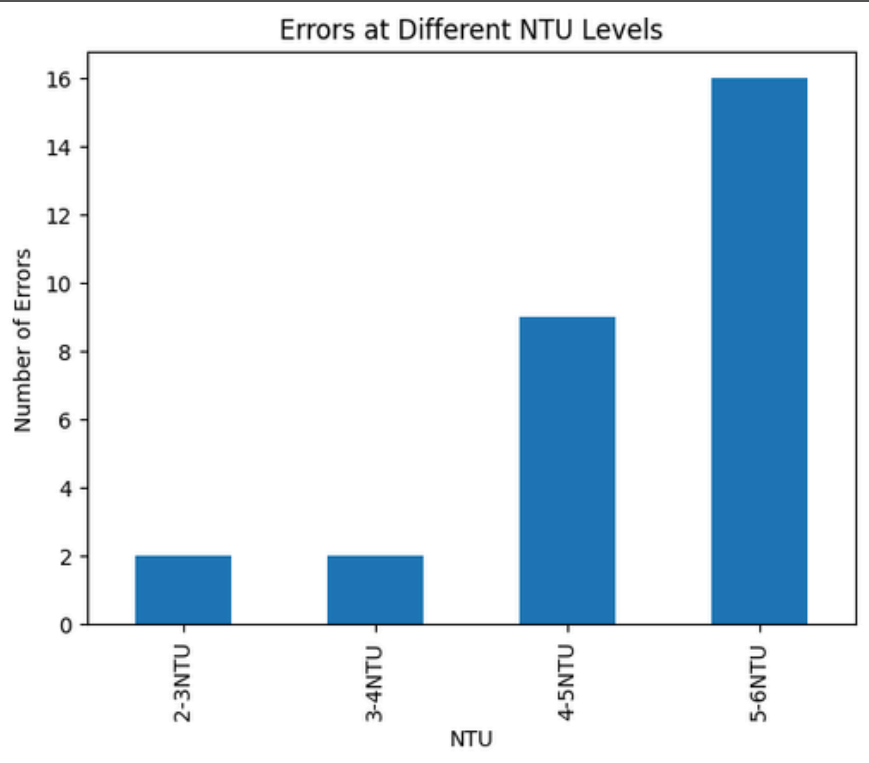


Day 25 - Detailed Error / Failure Analysis

Error Analysis Summary

Metric	Value
Total Test Samples	120
Correct Predictions	91
Misclassified Samples	29
Accuracy	75.83%

NTU-wise Error Distribution



NTU Level	Number of Errors
2-3 NTU	2
3-4 NTU	2
4-5 NTU	9
5-6 NTU	16

Key Findings :

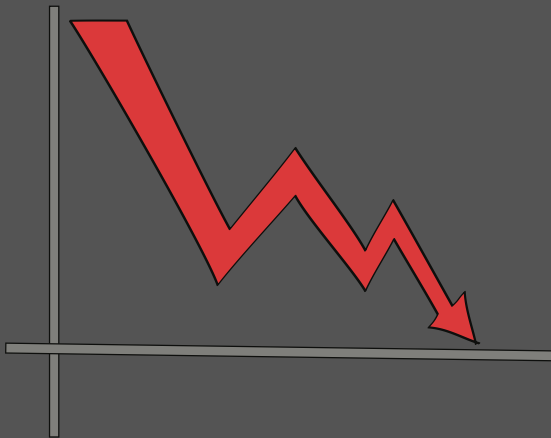
- Most prediction errors occurred at higher turbidity levels (4–6 NTU).
- Error count increased significantly as NTU increased.
- Low turbidity conditions (2–4 NTU) showed very few classification errors.
- High turbidity causes greater optical signal attenuation and scattering, leading to overlap between bit 0 and bit 1 signal values.
- The model performs reliably under low and moderate turbidity but struggles under severe channel degradation.

Failure Patterns Identified :

1. **High NTU Conditions:** Increased scattering and absorption distort the received optical signal.
2. **Signal Overlap:** Analog values of transmitted bit 0 and bit 1 become less separable.
3. **Limited Dataset Size:** Small dataset size restricts model generalization under challenging conditions.

Limitations :

- Dataset contains only 399 samples.
- Moderate class imbalance (60.65% bit 0 vs 39.35% bit 1).
- Limited feature set (Analog Value, Analog Voltage, NTU).
- Performance decreases as turbidity increases.



Thank
You

